CODING AND COFFEE

Phase 1.1: Hello, World!

Phase Goal

This phase will introduce you to creating, writing, and running Java programs in Eclipse. In this program, we'll be printing the following to the console:

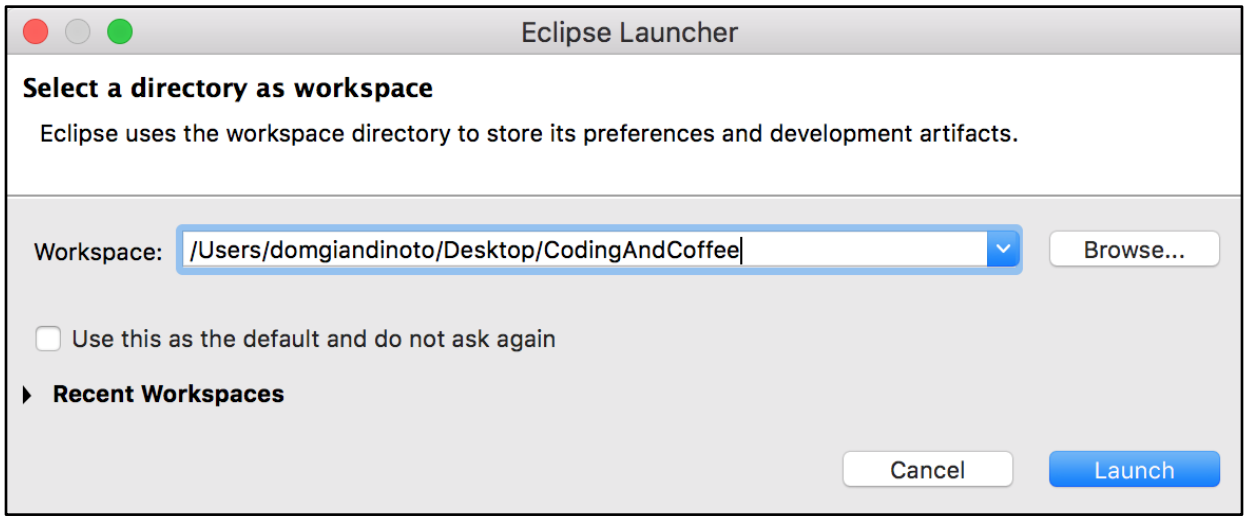
```
Hello, World!
This is [your name]!
```

"Printing to the console" just means that when you run your program in Eclipse, you'll see the above text on the "Console" tab at the bottom of the screen. Honestly a lot of things might seem confusing, but just trust us and trust our process. Everything will make more sense later, don't worry about it now!

Let's get started:

STEP 0: Creating a Java Workspace

In order to create a program in Java, we must first create a **Project**, or a folder to hold the program and all the things that Eclipse creates to go with it. You'll only have to do this once.

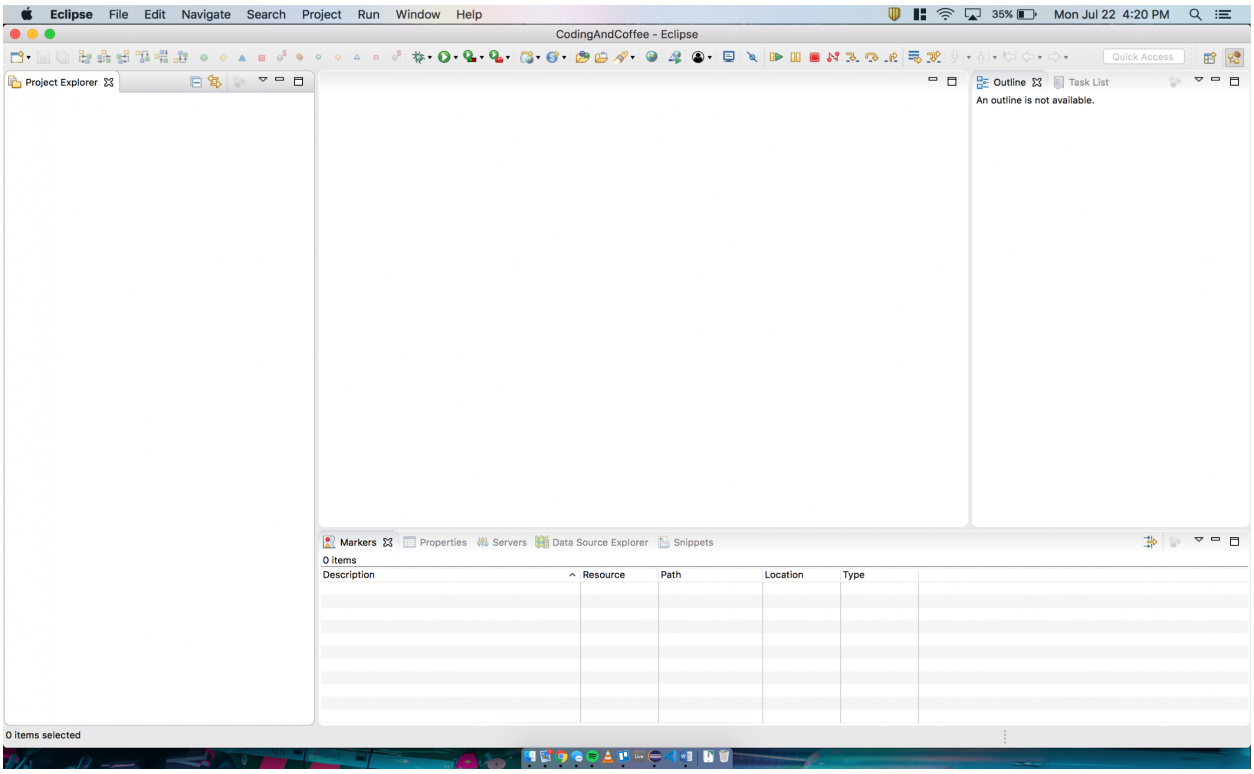


When you open Eclipse, you will see a prompt to open a **workspace**. A workspace is just a folder to keep your projects. For example, all of your Coding and Coffee projects can live in one "workspace."

Once you've named your workspace *(example above)* you can click "Launch."
NOTE: You'll see a welcome screen, but you can just close that by clicking the "X" in the tab in the top-left.

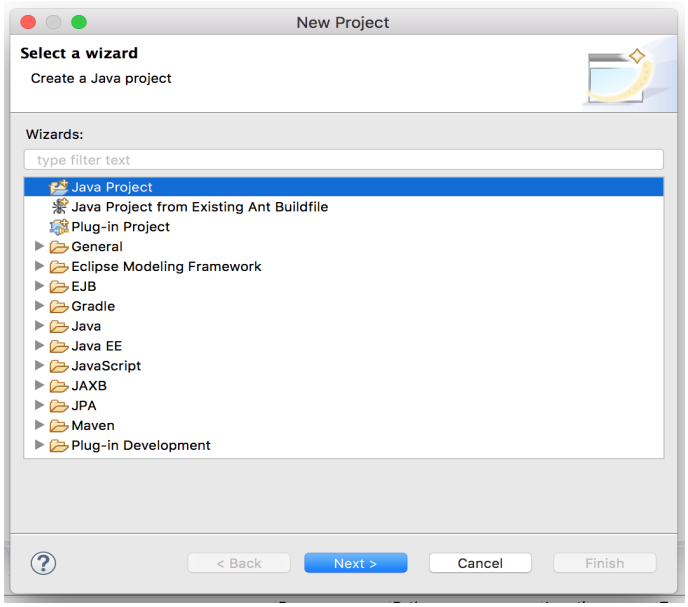
STEP 1: Creating a Java Project

Now you should see a screen like this:

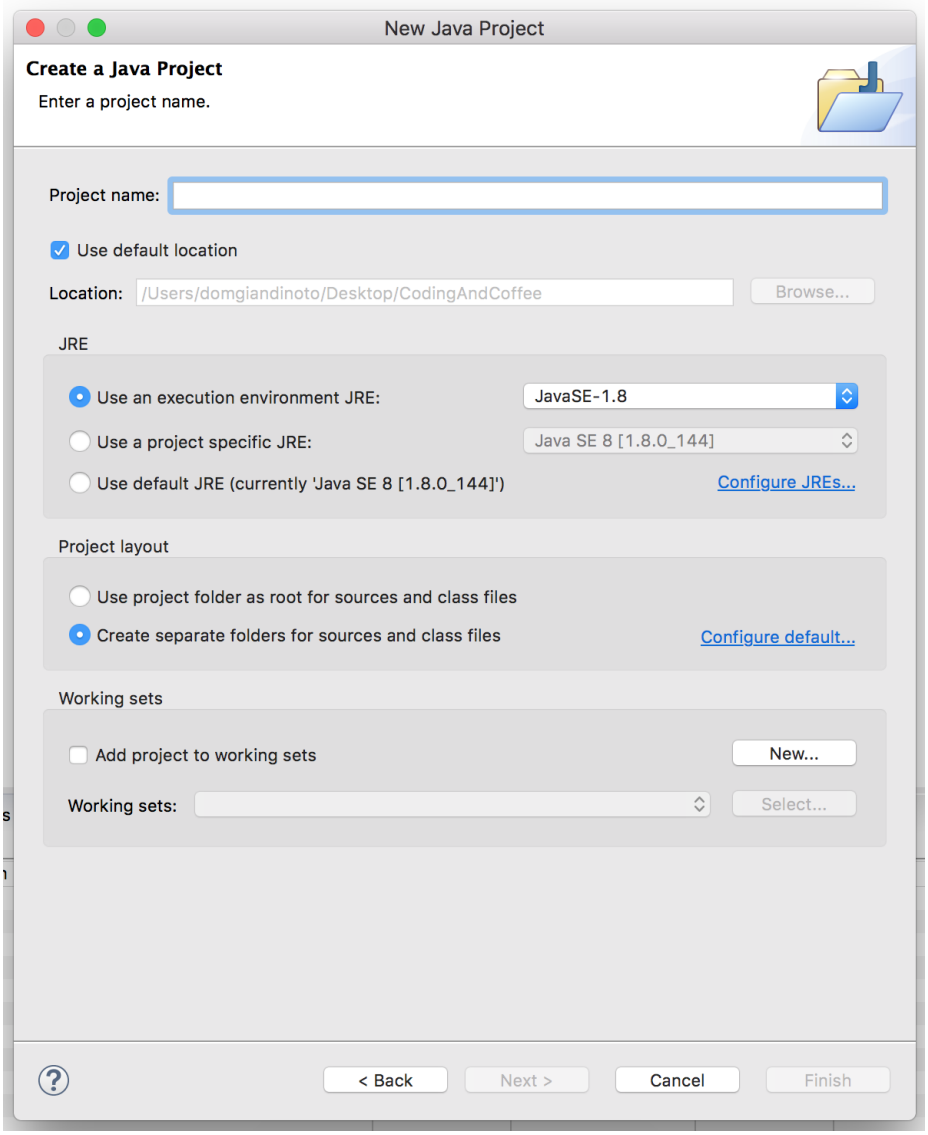


To create your Project, go to the top left and click File > New > Project. You should see a screen like this:

Then you should see a screen like this:



From here, you can select “Java Project” and then click “Next.”



Here is where you can name your project. We can just name this one “Phase1” for all your Phase 1 programs. After typing “Phase1” in the “Project name” box, type “Phase1_HelloWorld.” Then you can click “Finish.”

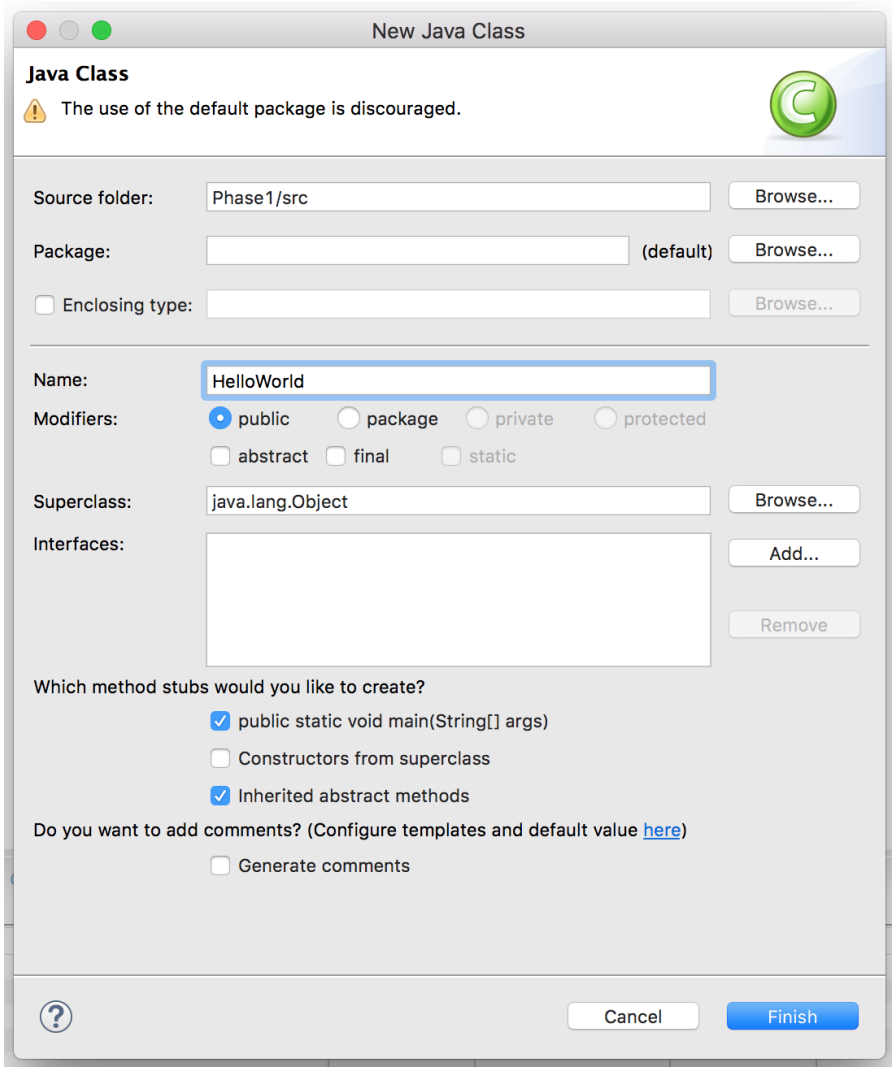
If a prompt comes up that asks if you want to open the Java Perspective, go ahead and agree to that.

STEP 2: Creating your program (AKA class)

Now it’s time to create the file where you’ll write your program. In Java, these files end in “.java” and contain **classes**. A **class** is a concept that we’ll go deeper into later, but for now,

just know that everything in a ".java" file is a class or is contained inside a class. You can refer to is as a file if it makes it easier.

To create your file, simply navigate to the File > New > Class. You'll see a window like this:



Make sure your source folder is "Phase1/src" and that the "public static void main(String args[])" box is checked. Name your class "HelloWorld" and click "Finish."

STEP 3: Write your code!

Boof. That was a lot! Don't worry, the first time is weird but after a while it will become second nature.

Now that we've got our class (or file) created, we can finally write some code! This program is really simple; we actually only need to write a couple lines. Make your program look exactly like this:

```
public class HelloWorld {
    public static void main(String args[]) {
        // your code goes here
        System.out.println("Hello, world!");
        System.out.println("This is Dom Giandinoto!");
    }
}
```

The `System.out.println()` lines that we wrote simply print the text to the console (minus the quotes) when you run your program. (It's kind of cryptic right now, but we'll get into how it all works later, just trust us.)

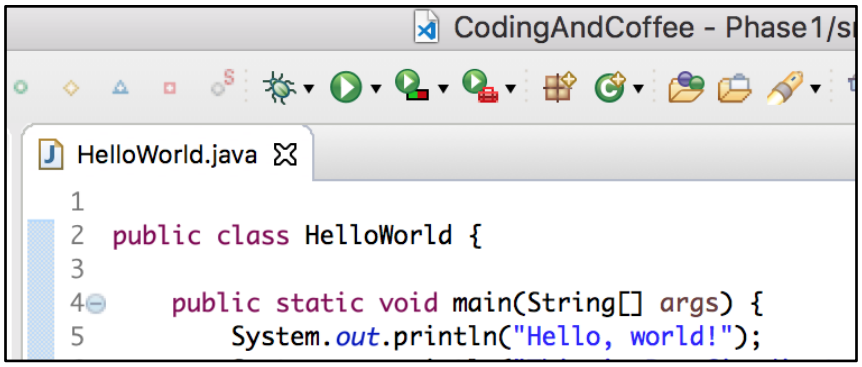
The line beginning with the two slashes (`//`) is called a **single-line comment**. Comments are a very important concept in any programming language. Whatever you write within a comment does not affect the outcome of your program at all; the computer will just ignore it. Comments don't have to be just one line either: multi-line comments in Java look like this, and will be ignored all the same.

```
/**
 * Hey there, this is a
 * multi-line comment.
 *
 */
public class HelloWorld {
    public static void main(String args[]) {
    // your code goes here
    }
}
```

You should always include some comments in your program, as they can tell others (or you) what your lines of code are doing.

STEP 4: Run your program

To run your program, simply click the green play button in the toolbar at the top of the screen.



Once you do that, you'll see your output in the "Console" tab at the bottom of the window.

Congratulations! You've made your first Java program! Now you can move on to Phase 1.2, where you'll learn about variables and operators.

<u>If you don't remember much</u>, remember that you ALWAYS need the lines below before you start coding in Java.

```
public class HelloWorld {
    public static void main(String args[]) {
      // your code goes here!!
    }
}
```

It won't always be "`HelloWorld`" (the name of your program) but everything else will be the same.

Also, sometimes a lot of background noise can make it <u>hard to focus</u>. The best programmers are just the ones who can focus the best. But some, like me, wear bulky, noise-cancelling headphones. If you had a hard time focusing, maybe try throwing on some headphones or earphones. For me, some music and a cup of coffee is a **must** when I sit down to code. I can't code without either of those things.